Persistent Semantic Scaffolds and the Pay-It-Forward Record: Reconstruction Cost and Reasoning Lineage in Long-Horizon Human-AI Systems

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Abstract

This paper introduces two architectural concepts for sustaining reasoning continuity in long-horizon institutional and human-AI systems: the Persistent Semantic Scaffold (PSS), a structured substrate for containing semantic mutation as systems grow in complexity, and the Pay-It-Forward Record (PIFR), a minimal artifact for carrying reasoning lineage forward across reasoning vehicles. Building on Paper 3's account of Interpretive Entropy (Mayoh, 2026a) and Paper 4's account of the Continuity Threshold (Mayoh, 2026b), this paper formalises the mechanism by which growing systems become harder to reconstruct — what it terms Semantic Surface Area and Mutation Blast Radius — and proposes structural responses grounded in organisational memory theory, distributed cognition, and software architecture erosion research. The argument is presented as a navigational hypothesis: the central claim is that beyond certain thresholds of complexity and institutional duration, structured reasoning scaffolding shifts from optional design preference to structural requirement. Neither PSS nor PIFR are proposed as software products or governance ideology, but as candidate layers of reasoning infrastructure for systems — human, institutional, or AI — that must remain coherent across long stretches of time.

Keywords: semantic scaffolds, reasoning lineage, reconstruction cost, organisational memory, distributed cognition, navigational coherence

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I. The Problem of Growing Without Forgetting

Every long-lived institution faces a quiet contradiction. To remain useful, it must grow — accumulate decisions, definitions, precedents, relationships, and accommodations to changing circumstances. But every addition makes the system as a whole harder to fully hold in mind. What began as a small, legible set of shared understandings becomes, over years, a sprawling structure that no single participant — human or otherwise — can entirely reconstruct from first principles.

This is not a failure of discipline or a problem solvable by better documentation alone. It is a structural consequence of growth itself, and it has a parallel that has been studied carefully in an adjacent field for half a century.

Software engineering research on system evolution has long observed that as a system evolves, its complexity increases unless deliberate work is done to reduce it — a dynamic researchers describe explicitly in terms of entropy, where change introduces disorder and structure degrades unless something actively pushes back. The underlying cause is rarely a single bad decision. It is that the information needed to understand a system's current state becomes scattered across too many files, too many contributors, and too many moments in time for any single mind to hold.

The parallel to institutional and reasoning systems generally is direct. What software architecture research calls structural decay, this research series has called Interpretive Entropy: the gradual divergence between original intent and current operational interpretation, accumulating not through any single failure but through the ordinary, unavoidable pressure of growth over time.

This paper asks what kind of infrastructure could meaningfully resist that pressure — not eliminate it, since Paper 3 already established that some entropy is structurally guaranteed (Mayoh, 2026a), but contain it well enough that institutions can continue growing without periodically requiring expensive, lossy reconstruction of their own reasoning.

II. Semantic Surface Area

To reason about this problem precisely, it helps to name what actually grows.

Semantic Surface Area (SSA) is the total set of semantic anchors, cross-references, governance constraints, and operational dependencies that define an institutional system. It is not merely the volume of documentation a system produces — a large archive of unrelated, self-contained records has comparatively low semantic surface area, because little of it depends on anything else. SSA grows specifically when definitions, terms, and decisions become entangled with one another: when a single concept is referenced across many decisions, when a constraint defined early shapes choices made much later, when the meaning of a term has accumulated a long, branching history of refinement and exception.

This entanglement is precisely what the software architecture literature studies under the heading of dependency propagation. Research on architectural technical debt has found that software systems are typically composed of interdependent components, and that the propagation of structural debt through those dependencies leads to complex, cascading effects that are exceptionally difficult to predict from any single component's perspective (Yang, Verma, & Anton, 2023).

As SSA increases, three things happen reliably. Mutation propagation pathways multiply — a change to one anchor point increasingly affects others not obviously connected to it. Reconstruction cost increases — recovering the reasoning behind any single decision increasingly requires understanding the web of dependencies around it, not just the decision in isolation. And drift probability increases — with more entangled anchors and more participants touching them over time, the likelihood that some local reinterpretation silently diverges from original intent grows accordingly.

None of this is a claim that growth is bad. It is a claim that growth has a structural cost that conventional documentation practices do not account for, because they typically measure volume rather than entanglement.

III. Reconstruction Cost

Paper 5 of this series, Invisible Opportunity Cost, established that the value lost through unnecessary reconstruction of understanding is structurally invisible to current economic measurement (Mayoh, 2026, forthcoming). This paper proposes a more specific, system-level companion concept.

Reconstruction Cost (RC) is the cognitive and procedural effort required to reconstruct the reasoning lineage that produced a given institutional state — not the state itself, but the chain of intent, assumption, and constraint that led to it.

RC increases as semantic surface area expands, as the justifications behind decisions fragment across multiple sources, as cross-references multiply, and as the historical context behind any given anchor deepens. This is not a hypothesis unique to this paper. The organisational memory literature, beginning with Walsh and Ungson's influential 1991 framework, has long observed that organisations struggle to articulate where institutional memory is actually stored, and that this difficulty compounds as organisations age and restructure.

Edwin Hutchins' distributed cognition theory offers the clearest account of why reconstruction becomes so expensive once context is lost. Hutchins argued that mental representations are not confined to individual minds but are distributed across people, artifacts, and the cultural systems through which a group interprets its shared work. Applied to institutional reasoning, this means the original "understanding" behind a decision was very often never held by any single person in the first place — it existed in the relationship between several participants and the tools and conventions they shared at the time. When those participants disperse and those conventions shift, there is no single mind left to query. The reasoning must be reconstructed from fragments, by people who were never part of the system that originally held it together.

This is also why reconstruction is not merely effortful but lossy. Michael Polanyi's foundational observation that we can know more than we can tell applies here directly: a substantial portion of the reasoning behind any institutional decision was tacit at the time it was made, embedded in shared assumptions that nobody thought to write down because they did not yet seem worth stating. Reconstruction efforts inevitably recover the explicit residue of a decision while losing much of the tacit scaffolding that made it coherent.

When Reconstruction Cost exceeds what participants can practically absorb, three predictable behaviours follow: defensive simplification increases, as participants narrow their engagement to avoid the cost of full reconstruction; the temptation to simply rewrite rather than understand and extend prior work rises; and silent reinterpretation becomes attractive, because quietly redefining a term locally is cheaper than tracing its full lineage.

IV. Mutation Blast Radius

Not all changes to an institutional system carry equal risk, and the difference is rarely obvious from the size of the change itself.

Mutation Blast Radius (MBR) is the extent to which a proposed semantic or structural change propagates across dependent anchors within the system's semantic surface area. A change that looks small in isolation — adjusting the definition of a single term, narrowing the scope of a single constraint — can carry an extremely high blast radius if that term or constraint functions as a hub that many other parts of the system depend on.

This is, again, a phenomenon already studied carefully in software architecture research, where it goes by the name of breaking change propagation. Engineering practice around microservice architectures has repeatedly documented that asynchronous changes to a shared interface can lead to breaking compatibility and systemic instability across services that depend on it — instability that is difficult to predict in advance without a formal map of dependencies. The discipline of semantic versioning exists specifically because engineers learned, often the hard way, that a change's apparent size is a poor predictor of its actual impact once dependencies are accounted for.

High-MBR environments require a different kind of caution than low-MBR ones. Before a high-blast-radius change is made, a system needs some mechanism for inventorying what currently depends on the thing being changed, articulating clearly why the change is needed, evaluating whether the change preserves backward interpretability or breaks it, and securing validation from whoever holds the authority to approve a change of that scope.

Without such a mechanism, institutions tend toward one of two failure modes: either excessive caution, in which even minor, beneficial changes are avoided because nobody can confidently bound their blast radius, or excessive carelessness, in which changes are made locally without anyone realising how far their effects will travel.

V. Persistent Semantic Scaffolds

The concepts above describe a problem. This section proposes a structural response.

A **Persistent Semantic Scaffold (PSS)** is a structured, addressable, mutation-governed semantic substrate designed to anchor definitions explicitly, preserve reconstructable reasoning lineage, enable incremental rather than wholesale revision of meaning, support authority-scoped validation of high-blast-radius changes, and allow a system's semantic surface area to grow without losing the ability to inspect how it got there.

A PSS differs in kind from conventional version control systems, knowledge graphs, and policy engines, even though it shares surface features with all three. Version control systems track changes to files; a PSS is concerned with the stability of meaning itself, which can drift even while the underlying documents remain unchanged. Knowledge graphs represent relationships between entities; a PSS is additionally concerned with the reasoning lineage behind why a relationship was established, not merely that it exists. Policy engines enforce rules; a PSS is concerned with preserving the ability to understand why a rule exists in the first place, which enforcement alone does not guarantee.

The function a PSS performs has a name in the organisational literature already, though it is rarely framed in infrastructural terms: **Inspectable Accumulation (IA)** — the capacity of a system to increase in complexity while preserving reconstructable reasoning and semantic transparency. Without inspectable accumulation, systems oscillate between two unstable states: drift, in which meaning silently diverges from intent until nobody can say with confidence what a term or rule actually means anymore, and dogma, in which the system responds to the threat of drift by freezing interpretation entirely, sacrificing the system's ability to adapt in order to preserve the appearance of stability. With inspectable accumulation, repair becomes possible in place of either drift or dogma: a system can be corrected locally, with its reasoning lineage intact, rather than requiring a wholesale rewrite to restore coherence.

A structural mechanism this paper names the **Authority Membrane (AM)** is the boundary through which a proposed mutation must pass before it is permitted to alter canonical meaning. Its function is to separate advisory reasoning — proposals, drafts, exploratory revisions — from authoritative mutation, ensuring that high-blast-radius changes are evaluated for justification sufficiency, blast radius, and continuity preservation before they are allowed to take effect.

VI. The Pay-It-Forward Record

Where the Persistent Semantic Scaffold addresses the structure of an evolving system as a whole, the **Pay-It-Forward Record (PIFR)** addresses a narrower and more immediate problem: how does a single piece of reasoning survive the handoff from one reasoning vehicle to the next, human or AI, without requiring that the next vehicle reconstruct the original context from scratch?

A PIFR is the minimum structure required for reasoning to continue without reconstruction. It typically records intent, the assumptions the reasoning depended on, the reasoning steps themselves, the constraints that bounded the reasoning, the conclusions reached, and explicit criteria for how the reasoning should be continued or revisited in future. A further field, **criteria for success** — a statement, made at initiation, of what would need to be true for the reasoning to count as having worked — is recommended at initiation, though not required.

This structure is deliberately closer to what cybernetics calls a feedforward artifact than a conventional record of a decision. Norbert Wiener's foundational distinction in cybernetics separates feedback mechanisms, which correct deviations after they occur, from feedforward mechanisms, which carry forward the information needed to anticipate and guide future behaviour before a deviation occurs. A conventional log — a record of what happened — is feedback in form, even when it is meticulously detailed: it tells a future reader what was decided, but not how to extend that decision into circumstances the original reasoning never anticipated. A PIFR is structured instead to carry forward the elements of a working model of the original reasoning, so that a future participant can anticipate how that reasoning would extend, not merely retrieve what it once concluded.

This distinction matters because of a specific and well-documented failure mode in institutional knowledge transfer. The succession-planning and organisational learning literature has found repeatedly that organisations tend to document explicit, step-by-step process knowledge while overlooking the tacit reasoning that explains why key decisions were made the way they were — precisely the kind of context that future participants need most and are least likely to find written down. A PIFR is structured specifically to resist that failure mode: it does not document only the conclusion of a piece of reasoning, but the intent and assumptions that made the conclusion sensible in its original context.

PIFRs do not, by themselves, accumulate institutional knowledge — that is the role of the broader scaffold within which they sit. What they enable is something more specific: the succession of reasoning vehicles across time, such that a future participant — another team member, a later version of an institution's policy, or a different instance of an AI system — can pick up a line of reasoning without first having to excavate it.

VII. Institutional Longevity as a Distinct Design Criterion

Systems built to support reasoning continuity over long horizons face constraints that are easy to overlook when they are designed using the assumptions and incentives of conventional product architecture.

Conventional software and institutional tooling is frequently optimised for velocity: shipping features quickly, iterating rapidly, treating yesterday's decisions as provisional and easily superseded. This is a reasonable orientation for systems whose primary value lies in immediate functionality. It is a poor orientation for infrastructure whose primary value lies in preserving the interpretability of decisions made years, or generations, earlier.

Semantic scaffolds supporting institutional continuity need to be evaluated against a different set of criteria: backward interpretability, meaning a participant encountering an old decision today can still reconstruct why it was made; mutation containment, meaning changes do not propagate further than their justification warrants; continuity of meaning across personnel and platform changes; and stability under temporal extension, meaning the system's usefulness does not degrade simply because more time has passed.

This is not an argument against velocity in general. It is an argument that reasoning infrastructure intended to last across long horizons of institutional time needs to be designed and evaluated on its own terms, rather than inheriting evaluation criteria built for faster-moving, more disposable systems.

VIII. Limitations

Several limitations apply directly to the proposals in this paper, and are worth stating plainly rather than implying a completeness the framework does not yet have.

Persistent semantic scaffolds do not eliminate interpretive entropy; Paper 3 of this series already established that some entropy is a structural feature of fragmented systems, not an incidental failure correctable by sufficiently good tooling. Scaffolding can slow and contain entropy's accumulation; it cannot prevent it outright. Disciplined mutation protocols are required for the Authority Membrane concept to function as intended — the membrane is only as effective as the rigor with which proposed changes are actually evaluated before being allowed through it. Additional structural overhead is introduced by any scaffold of this kind, and that overhead needs to be weighed honestly against the reconstruction cost it is meant to reduce; a scaffold that is more expensive to maintain than the reconstruction it prevents has not solved the underlying problem. And the threshold conditions under which scaffolding moves from optional to structurally necessary — the central hypothesis of this paper — remain a matter for empirical investigation rather than established fact.

IX. Research Agenda

This paper presents a conceptual architecture grounded in established literature from organisational memory, distributed cognition, and software architecture research, rather than controlled empirical findings of its own. Several lines of future investigation would be needed to test its central claims directly: comparative measurement of drift in scaffolded versus non-scaffolded institutional systems over matched time horizons; empirical measurement of reconstruction cost as semantic surface area scales, to test whether the relationship is linear or — as the Continuity Threshold hypothesis from Paper 4 would predict — exhibits threshold dynamics; analysis of mutation blast radius prediction accuracy, testing whether blast radius can be estimated in advance with useful precision; and study of continuity outcomes specifically across AI instance turnover, where the reasoning vehicle changes entirely rather than merely the human personnel involved.

X. Conclusion

This paper has argued that interpretive entropy is not an incidental failure of institutional discipline but a structural consequence of growth in any system that accumulates semantic surface area over time — a dynamic already documented carefully, under different names, in organisational memory research and in the study of software architecture erosion.

It has proposed Persistent Semantic Scaffolds as a candidate structural response: an infrastructure layer designed to make growth in institutional complexity inspectable rather than opaque, governing high-blast-radius mutation through an explicit authority boundary rather than leaving it to informal convention. And it has proposed the Pay-It-Forward Record as the minimal artifact through which a single piece of reasoning survives the handoff between reasoning vehicles — carrying forward intent and assumption, not merely conclusion, in keeping with the feedforward principle from cybernetics rather than the purely after-the-fact logic of conventional documentation.

The central hypothesis offered here is structural, not yet empirical: that beyond certain thresholds of semantic surface area and institutional duration, unaided reconstructive cognition becomes insufficient to maintain interpretive coherence, and that mutation containment and reasoning preservation mechanisms accordingly shift from optional design preference to structural requirement.

Neither PSS nor PIFR are proposed here as finished software products or governance doctrine. They are offered as a candidate layer of reasoning infrastructure — a starting hypothesis for what long-horizon human-AI systems may require if they are to grow without losing the capacity to explain themselves.

Declarations

Author Affiliation and Research Programme

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Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

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